

Dynamic Capabilities and the Speed of Strategic Change: Evidence From China

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Abstract—Despite the general agreement of the importance of resources and capabilities to strategic change, the broader question of how resources and capabilities affect strategic change itself remains unresolved. This research focuses on one characteristic of the process of strategic change, i.e., its speed. This study investigates the relationship between dynamic capabilities, the speed of strategic change, and performance. We propose that technological capabilities, marketing capabilities, and market-linking capabilities, due to their absorptive capacity and boundary spanning attributes, will facilitate the speed of strategic change. Furthermore, the risk orientation of managers will moderate this effect such that it would be strengthened for relatively risk-taking managers. Finally, we argue that the speed of strategic change will have an inverted-U relationship with organizational performance. We test these hypotheses on a sample of 213 Chinese firms and find that while dynamic capabilities facilitate rapid strategic change, the risk orientation of decision makers moderates this relationship, but not all as hypothesized. While faster implementation of strategic change has a positive effect on performance, strategic change carried out too quickly has negative performance repercussions. Our findings suggest that firms not only need to adjust strategies to complement the changing environment, but also have corresponding capabilities to enable rapid strategic change.

Index Terms—Firm performance, marketing capabilities, market-linking capabilities, risk orientation, speed of strategic change, technological capabilities.

I. INTRODUCTION

STRATEGIC management is concerned with environmental changes and organizational adaptation [1]. Organizations adapt to their environment through their strategic changes. Consequently, strategic change is necessary and important, especially in dynamic environments [2]–[4], and the ability to make appropriate strategic changes quickly has become a potential source of competitive advantage to firms [5]–[10]. However, despite this expectation, empirical investigations into the effect of strategic change on performance have proven equivocal [10]. While some studies have found that strategic change positively

impacts performance [11], other studies found negative effects [5], [12]. Moreover, some studies found mixed [13] or no effect [2] of strategic change on performance. These results indicate that the nature of strategic change and its effect on firm performance maybe more nuanced and complex [14].

Strategic change is defined as “a difference in the form, quality, or state over time” [15]. Embodied in this definition are three dimensions of change: the type of change, the magnitude of change (both captured by “form” and “quality”) and the speed of change (captured by “over time”). While prior strategic change research has examined elements of the type and magnitude of change [16]–[18], inadequate research has focused on the speed of change. Yet the speed of strategic change can also affect performance; indeed, the competitive global economy necessitates that firms learn, innovate, and take action ever faster [5]. The faster strategic change can be implemented, the more opportunities a firm can seize prior to competitors and thus gain an advantage [19], [20]. The speed with which a specific strategic change is implemented reflects not only the urgency and priority managers place on the change, but also the capabilities the firm has to underwrite the change process. Thus, the speed of strategic change is potentially an important dimension that can illuminate our understanding of strategic change and its effect on firm performance.

In this paper, we propose and test a theory that links dynamic capabilities to the speed of strategic change and subsequent performance. A firm’s ability to make relevant changes to its strategy, and to do so quickly, is predicated on the resources and capabilities it possesses [2]. Capabilities provide the slack and flexibility necessary to quickly adjust to environmental changes and therefore will facilitate fast strategic change [21], [22]. We further propose that the relationship between dynamic capabilities and the speed of strategic change is moderated by the risk orientation of management; this risk orientation of decision makers critically influences not only the decision-making process [23], [24] but the awareness and speed with which executives respond to organizational challenges as well [25]. Finally, we argue that the speed of strategic change has an inverted “U” relationship with performance such that firms that carry out strategic change too quickly, or suffer from inertia, underperform. We test these relationships using a sample of 213 Chinese firms.

This research contributes to the strategic change literature by investigating a hitherto underexamined dimension of strategic change, i.e., the speed of strategic change. Most prior research investigating strategic change has focused on two aspects: 1) the type of change; and 2) the magnitude of change. While there has been limited research on the speed of strategic change, earlier studies have examined component processes,

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such as strategic decision-making speed (e.g., [7], [26]) and competitive action and response speeds (e.g., [19]). No paper to date, to the best of our knowledge, has examined the speed of the entire strategic change process, from decision-making to action implementation. We find that a moderate speed of strategic change can best benefit firm performance. Thus, a better understanding of the speed of strategic change further reveals nuances that lead to conflicting findings on the effect of strategic change on performance.

Furthermore, despite the salience of the speed of strategic change in theoretical characterizations of the effects of dynamic capabilities, there is little empirical investigation about how dynamic capabilities actually influence the speed of strategic change. While prior research has examined the effect of dynamic capabilities on the magnitude of strategic change, no such attention has been dedicated to the speed of strategic change. This paper also contributes to the literature by addressing this oversight in examining how dynamic capabilities influence the speed of strategic change.

Finally, while prior research has investigated direct effects of decision makers' characteristics on strategic change [14], [27], the role of decision makers in the capability-strategic change relationship has not been identified. Consistent with recent advances in resource management research that argues for the central role of managers in converting firm resources and capabilities into competitive action [28], we hypothesize and empirically test how firms' managerial risk orientation moderates the relationship between dynamic capabilities and the speed of strategic change. Thus, the findings of this study provide new evidence to the role of management in realizing the value of firm resources and capabilities through the pace of strategic change.

II. THEORY AND HYPOTHESES

The central role of dynamic capabilities is facilitating change within an organization [29]. Indeed, Helfat *et al.* argue that the essence of dynamic capabilities lie in identifying the need for change within firms, formulating a response to this need, and implementing the chosen course of action [27]. While a firm might face the need to change, it may not have the ability to do so, thus delaying necessary strategic change [11]. Therefore, how quickly a firm identifies, formulates, and implements change is dependent on its dynamic capabilities. Yet, despite the salience of the speed of strategic change in the dynamic capabilities literature, scant empirical research to date has examined this linkage. Furthermore, although researchers have examined the effects of subcomponents of the speed of strategic change—i.e., decision-making speed [7] and the speed of action or response [30]—little research has examined the effect of speed of the entire strategic change process. We, therefore, investigate the effect of dynamic capabilities on the speed of strategic change and the subsequent effect of the speed of strategic change on firm performance. Change as used in the literature encompasses strategic and tactical change. Both strategic change and tactical change can create value for shareholders but differ in their influence. Strategic change helps firms improve their capabilities, while tactical change only enables direct influence on short-

term value [31]. In this paper, we refer only to strategic change because our focus is on dynamic capabilities, and our research context is a dynamic environment in which strategic change is crucial [2]. Our theoretical framework is summarized in Fig. 1.

A. Dynamic Capabilities and the Speed of Strategic Change

The speed of strategic change reflects the time taken to change strategies, which consists of the decision-making speed [7] and the speed to implement new strategies [6]. Strategic decision-making speed refers to the length of time between initiating and formulating a strategic decision, usually starting from the first time someone within the firm proposes a meeting or seeks information to the time the firm is ready to implement a well-developed strategic decision [7]. Implementation speed refers to the length of time between the promise to implement and the integration of the strategic decision with operations [6]. We conceptualize the speed of strategic change as the time spent on the process of designing and/or developing and implementing strategic change [5]–[7].

A firm's ability to quickly change its strategy is driven, among other things, by the capabilities it possesses [32]. Competitive dynamics research has identified awareness, motivation and capabilities (the AMC framework) as the key drivers of a firm's ability to initiate competitive action or respond to competitors [30]. While awareness addresses the need for change, and motivation the reasons for change, the ability to act or change is driven by firm capabilities [28]. A firm's capabilities provide it with "... a bundle of options for future strategic choice" [33, p. 762] with which it can quickly change strategy. Without the requisite capabilities, a firm that is aware of the need to change, and motivated to do so, will be unable to implement such change in a timely manner. It is, therefore, unsurprising that competitive dynamics research has found a firm's capabilities to drive the actions it can take [28] and how quickly it can respond to competitor actions [30].

While capabilities in general fuel a firm's actions, dynamic capabilities determine how quickly they can formulate and implement actions. Dynamic capabilities are defined as "the organizational and strategic routines by which firms achieve new resource configurations..." [34]. Teece *et al.* [35] for example conceptualized dynamic capabilities as capabilities needed for quick organizational adaptation in fast-changing environments. They enable a firm to quickly reconfigure its organizational routines. Central to the characterization of dynamic capabilities is not only the strategic flexibility afforded to firms [36], but the speed with which they enable management to redesign and reconfigure organizational structure and routines necessary for change [37]. Firms that possess dynamic capabilities can quickly reconfigure other capabilities and organizational routines and thus implement strategic change quickly. Thus, dynamic capabilities facilitate the speed with which firms can implement strategic change.

We, therefore, propose hypotheses that link three central dynamic capabilities to the speed of strategic change. Specifically, we examine how technological capabilities, marketing capabilities, and market-linking capabilities facilitate the speed of

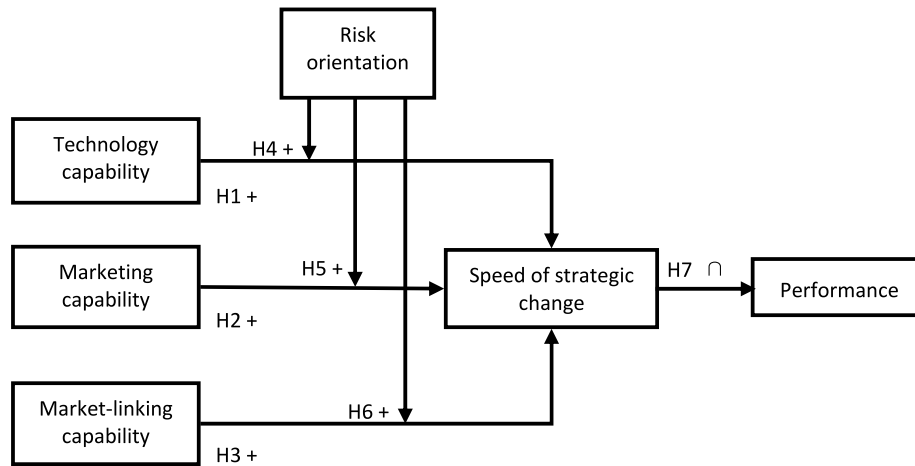


Fig. 1. Conceptual model.

strategic change. While the competitive dynamics and dynamic capabilities literatures elucidate why these dynamic capabilities speed strategic change, how they do so differs relative to the three capabilities. We discuss *how* each of these capabilities speed strategic change below.

1) Technological Capabilities and the Speed of Strategic Change: Technological capabilities represent a firm's ability to employ various technologies [38] and encapsulate the production skills and technological know-how embedded in the process of R&D and production [39]. They include technical resources, such as design and process skills and quality control programs, and capture how a firm aligns different types of production skills and integrates different types of cross-functional technological flows to design and produce products and services [40].

The literature provides two potentially conflicting effects of technological capabilities on the speed of strategic change. First, due to its path-dependence, technological capabilities are argued to hinder fast strategic change. Large investments in technological capabilities become sunk costs with changes in the environment, such as the invention of new technology. Irreversibility of technology investments drives up the costs to redeploy or give up those investments essentially making them sticky resources [41]. This entrenches the firm to a committed strategy and reduces the ability of the firm to develop new resources needed for entrepreneurial actions and strategic change. Such a tendency is echoed in organizational phenomena such as competency traps [42], core competencies becoming core rigidities, and the organization's focus on "static efficiency" to the detriment of "dynamic efficiency" [43].

Conversely, technological capabilities could also provide a firm with the absorptive capacity needed to quickly comprehend and respond to environmental changes and thus facilitate quick strategic change. Greater technological resources that span technological boundaries are essential for innovation [44], making it easier for a firm with more technological resources to quickly respond to changes in the environment. A wider breadth of technological resources allows a firm to formulate and implement a broader variety of actions [28], ensuring that the firm has

among its options requisite actions to react to environmental changes quickly. Technological capabilities enable the firm to acquire and synthesize knowledge across technological boundaries and thus reconfigure its resources to more quickly respond to changes in the environment. This combinative capability [45] or architectural competence [46] is central to providing firms with strategic flexibility. The more strategic flexibility a firm has, the faster it can implement strategic change [47].

We argue that in a fast-changing and dynamic environment, such as that faced by firms in this sample, the absorptive capacity effect dominates the path-dependence effect such that technological resources will facilitate rather than hinder the speed of strategic change. In quickly changing environments, the ability of firms to seek and acquire new external knowledge is central to avoiding myopic tendencies that induce inertia [48]. Indeed, Cohen and Levinthal [49, p. 122] conceptualized absorptive capacity as "... the organization's ability to recognize new external knowledge, assimilate it, and apply it to commercial ends." This ability to access and process new external knowledge is especially crucial for firms in dynamic environments that have to deal with ever changing variables [50]. The more technological capabilities a firm has, the greater its absorptive capacity and therefore the greater its ability to recognize, assimilate, and apply new external knowledge necessary for fast strategic change. Thus, we propose:

Hypothesis 1: Technological capabilities will positively relate to the speed of strategic change.

For a firm to initiate and implement fast strategic change, it needs to be able to redeploy its resources and capabilities quickly. Boundary spanning capabilities are those that link an organization with its environment [51]. These capabilities are crucial to strategic change because of their role in identifying and dealing with trends and changes in the firm's environment [51]. Boundary spanning enables organizations to move beyond local search behaviors to engage in further exploration [52]. Although local search behavior creates efficiency in new knowledge generation, it nonetheless constrains the firm's resource generation activities to only path-dependent exploration activities. Boundary spanning capabilities, however, enable a

firm to overcome local search tendencies to seek new external knowledge. We focus on the two boundary spanning capabilities: marketing capabilities and market-linking capabilities.

2) *Marketing Capabilities and Speed of Strategic Change*: Marketing capabilities are those developed to adapt to market changes. They include “knowledge of the competition and of customers and skill in segmenting and targeting markets, in advertising and pricing, and in integrating marketing activity” [45, p. 10]. As one type of boundary spanning capability, marketing capabilities facilitate faster strategic change for several reasons. First, marketing capabilities help a firm adapt to environmental changes [34], during which the most valuable information comes from competitors and customers [50]. This is exactly the knowledge that marketing capabilities produce and utilize. Once the knowledge of competitors and customers changes, firms initiate and implement corresponding strategic changes such as adjustments to advertising and pricing.

Second, the process of transferring marketing resources to products and services facilitates faster strategic change [31]. Marketing capabilities integrate cross-functional business knowledge to create and ensure fast delivery of customer value [34]. Indeed, the hallmark of effective marketing capabilities is the speed of responsiveness to customer changes [53]. Changes in customer needs and preferences are key drivers of environmental change [4]. To the extent marketing capabilities ensure fast responsiveness to customer changes, they facilitate the speed of strategic change.

Lastly, marketing capabilities emphasize actions to penetrate and create customers in new markets [54], which reflect proactive strategic adjustment. Successfully building new customer relationships results from successfully identifying and taking advantage of changes in customers’ needs and preferences. Marketing capabilities intertwine with the external environment and often directly serve the market, enabling continuous information flow and feedback. Timely and accurate assessment of information centered on environmental changes can help a firm correctly and rapidly judge the limitation of current strategies, decide on the direction and content of strategic change, as well as implement the agreed upon strategic changes. Essentially, firms with stronger marketing capabilities will be able to make quicker strategic changes. Thus:

Hypothesis 2: Marketing capabilities will be positively related to the speed of strategic change.

3) *Market-Linking Capabilities and Speed of Strategic Change*: Market-linking capabilities refer to those that enable the organization to sense market changes, create and maintain customer relations, and keep close contact with channel members such as wholesalers and retailers [44]. Those capabilities consist of market-sensing, customer linking, and channel integration capabilities [31]. Market-sensing capabilities enable the organization to observe market trends, sense market changes, and link other types of capabilities with the external environment [55]. This function is similar to that of marketing capabilities with respect to competitors and customers, but seeks information from more than competitors and customers. Hence, market-sensing capabilities can help the organization to effectively and

very quickly determine the necessity of strategic change, grasp market opportunities, and avoid market threats.

Customer-linking capabilities emphasize comprehensive gathering of external information, searching for market opportunities, exploring new markets, and creating market demand, thus linking the firm to potential customers. Once the firm needs to make strategic change, well-maintained customer relations can be a buffer to routines. More importantly, successfully linked new customer bases can be treated as experimental fields to implement strategic change, which will facilitate the outcome of strategic change [56].

Channel integration capabilities refer to those that integrate activities along the supply chain. Strong bonds with channel members can establish high consistency during the implementation of strategic change, improve the compatibility of strategic decisions among members, and guarantee the necessary resources from outside, as well as quickly delivering the value of strategic change [6].

Thus, market-linking capabilities, located at the boundaries of the organization, can obtain strategic support from environment [57] and quickly respond to the need for strategic change. It can build a strong network for the organization, which is critical in identifying opportunities, investigating feasibilities, and accumulating resources. The relationships with external organizations provide information about new technology and market opportunities for possible strategic collaboration [39]. Thus, market-linking capabilities facilitate the swift sensing of market changes, effective strategic decisions, and the fast transfer of those decisions to the network, and as such speedy implementation of strategic change. Therefore:

Hypothesis 3: Market-linking capabilities will be positively related to the speed of strategic change.

B. Moderating Effect of Managerial Risk Orientation

Recent research has highlighted the importance of management in leveraging a firm’s resources and capabilities into competitive action [58]. Managers’ cognitive frameworks have been argued to affect firm-level strategic behaviors [59]. Therefore, while the attributes of different capabilities determine how they facilitate the speed of strategic change, the risk orientation of the firm’s executives will influence those strategic decisions [60], their implementation [61], and thus whether or not a capability facilitates quick strategic change.

Risk orientation refers to the attitude of decision makers when they confront risky decisions, and includes three types: risk-taking, risk-neutral, or risk-averse. Since risk orientation reflects individual personality, it is stable, and thus has impact on decisions [23], [24]. With different types of risk orientation, decision makers will react differently to the same market opportunities, since they see different opportunities, feel different pressures, and respond to different drivers for actions.

Risk-taking decision makers like taking risks and enjoy the stimuli from previous risky experiences. Thus, facing market changes, they are likely to be eager to achieve success, exaggerate the attractiveness, proactively seek the potential benefits,

and thus generate the strong likelihood of actions [62]. Risk-taking decision makers may not comprehensively evaluate potential risks. With increasing capabilities, risk-taking managers may feel buffered and thus more aggressively pursue strategic change. In contrast, risk-averse decision makers are more conservative and try to avoid risks and potential failure. They will first evaluate the possibility of failure and try to avoid such failure. From their perspective, the decision maker will bear the responsibility of any failure [62]. With increasing capabilities, risk-averse managers will be more conservative and feel less pressure to implement strategic changes quickly. Therefore, we propose that risk orientation moderates the relationships between dynamic capabilities and the speed of strategic change.

Compared to risk-averse decision makers, risk-taking decision makers are more willing to give up current technological capabilities, fragment existing resource accumulation paths, and tackle resource inertia and capability rigidities. With more technological capabilities, risk-taking managers have the wherewithal to explore unknown areas and invest resources in uncertain conditions [63]. Such actions will create favorable conditions for new technology development, largely weaken the negative effects of technological path dependence, enhance the absorptive capacity effect of technological capabilities, and thus facilitate the speedy implementation of strategic change. In contrast, risk-averse decision makers will be more conservative with increasing technological capabilities, focusing more on exploitation (as opposed to exploration) of the capabilities, and thus exaggerate the negative effects of resource inertia and capability rigidities. Hence, the more risk-taking the decision maker, the stronger the positive relationship is between technological capabilities and the speed of strategic change. Thus:

Hypothesis 4: Risk orientation will positively moderate the relationship between technological capabilities and the speed of strategic change such that the positive relationship between technological capabilities and the speed of strategic change will be strengthened for firms with more risk-taking management and weakened for firms with more risk-averse management.

Risk-taking decision makers, once sensing a change in competitors and customers, can quickly grasp competitive opportunities and invest marketing capabilities to capitalize on the change [64]. The more marketing capabilities their firms have, the more likely they would seek to gain such first-mover advantages more quickly. Even if they cannot obtain first-mover advantages, they are able to follow the change in customers' needs or follow strategies of their strong competitors rapidly and thus divide benefits of first movers. In contrast, for risk-averse decision makers, more marketing capabilities may actually provide for information overload, slowing down decision-making while they digest the information, and thus slow the speed of strategic change. Hence, the more risk-taking a manager, the stronger the positive relationship is between marketing capabilities and the speed of strategic change. Thus, we propose:

Hypothesis 5: Risk orientation will moderate the relationship between marketing capabilities and the speed of strategic change such that the positive relationship between marketing ca-

pabilities and the speed of strategic change will be strengthened for firms with more risk-taking management and weakened for firms with more risk-averse management.

Compared to risk-averse decision makers, risk-taking decision makers proactively investigate the needs of customers, discover and retrieve potential new customer bases, and link potential customers with existing ones to establish new customer relationships. The more market-linking capabilities they have, the more willing they are to break the balance of power and be proactive in the supply chain [65], thus speeding strategic change. In contrast, risk-averse decision makers are often liable to maintain the existing customer base, slow the process, or even give up on the exploration of potential customers. The more market-linking capabilities they have, the more wedded they become to existing relationships and thus are reluctant to quickly implement change that may affect their stakeholder relationships. Hence, the more risk-taking a decision maker, the stronger the positive relationship is between market-linking capabilities and the speed of strategic change. Thus, we propose:

Hypothesis 6: Risk orientation will moderate the relationship between market-linking capabilities and the speed of strategic change such that the positive relationship between market-linking capabilities and the speed of strategic change will be strengthened for firms with more risk-taking management and weakened for firms with more risk-averse management.

C. Speed of Strategic Change and Firm Performance

In dynamic environments, firms experience frequent environmental changes. This challenges them to continuously adapt their strategies to stay competitive within the environment. Firms that can quickly adapt to these changes gain and sustain competitive advantages [7]. Previous studies suggest that fast decision-making in high-velocity environments leads to better performance [8]. Similarly, the speed with which decisions are implemented also affects performance. Firms that are aggressive in initiating competitive actions or quick in responding [19] perform better. First-mover advantages accrue to firms that are the fastest in adjusting to any change in the environment [20]. Therefore, the faster a firm formulates and implements strategic change, the more likely it is to reap performance benefits from the environment [5].

Although faster strategic change should improve performance, making changes too quickly may also be detrimental to performance. If a firm makes changes too quickly, it may sacrifice comprehensive information gathering and consensus building within management, yielding negative repercussions for performance [5]. Strategic changes made too quickly may lead to half-baked ideas implemented without full consideration of organizational effects. Therefore, although fast strategic changes will positively affect performance, strategic changes made too quickly will have the reverse effect. We, therefore, propose an inverted-U relationship between the speed of strategic change and firm performance.

Hypothesis 7: There will be an inverted-U relationship between the speed of strategic change and firm performance.

III. METHODS

A. Sample and Data Collection

China is an ideal empirical context within which to investigate strategic change due to the turbulent environment faced by firms operating there. The environment in the Chinese market is dynamic due to dramatic changes in policies, factor and consumer markets since the late 1970s. This was especially the case after China's entry into the WTO in 2001 [66].

We collected data through a survey. To reduce potential bias due to economic imbalance in different districts in China, we chose firms from the Pearl River Delta located in the southern part of China, the Yangtze River Delta in the east, and the Bohai Economic Rim in the north, and the Northwest Inland Regional Economic Zone in the mid-west. Our sample companies were randomly selected from registered corporations provided by the Economy Commerce Committee, a special administrative setup of the government for managing firms.

The questionnaire was originally designed in English based on the dynamic capabilities and strategic change literature. Four bilingual experts translated the questionnaire into Chinese. The Chinese version was subsequently back-translated into English by a third party to ensure accuracy [67]. Next, a pilot test was conducted with ten firms. During the process, interviewers checked each item with pretesters to make sure every question could be accurately understood. Respondents were top managers, which ensured that they were knowledgeable and possessed accurate information about their firms' strategic management practices. The survey was administered on site in each firm. The interviewers scheduled appointments to meet with the CEO and one other member of the TMT (e.g., CFO/COO/SVP) of each firm. Thus, there were two respondents per firm. A total of 650 firms were approached, and 232 firms participated. Due to missing data, our final sample includes 213 firms, which represents a response rate of 32.77%. To check for nonresponse bias, we compared the responding and nonresponding firms along firm attributes such as firm size, ownership status, sales, and age using the t-tests. All t-statistics were nonsignificant implying nonresponse bias is not an issue with our analysis. The measures used below are the average scores from the two respondents for each firm.

B. Measures

Performance was measured using four items that captured changes over the last five years in: 1) sales; 2) market share; 3) return on investment; and 4) net income for the focal firm. The respondents were asked to rate the level of change using Likert scales ranging from 1, "significant reduction" through 3, and "no change" to 5, "significant increase." Principal component analysis (PCA) indicated that all items loaded on a single factor, with an eigenvalue of 2.87, accounting for 71.71% of the variance and factor loadings ranging from 0.82 to 0.88. The Cronbach's coefficient alpha is 0.87.

Speed of strategic change was measured using four items modified from a scale used by Dooley *et al.* [6] and Kim and McIntosh [9]. The items include, relative to our major competitors.

- 1) We design strategic plans very quickly.

- 2) We implement strategic plans very quickly.
- 3) Our top managers agree with each other rapidly on design and implementation of new strategies.
- 4) Our employees accept firms' new strategies or strategic adjustment very quickly.

The interviewer instructed respondents to anchor their responses relative to competitors. We used the English adverb "very" because of the difficulty in translating the sentence "we design strategic plans quickly" into Chinese without using the Chinese adverb "very." Items were measured using Likert scales ranging from 1, "strongly disagree" to 5, "strongly agree." The PCA analysis indicated that all items loaded on a single factor, with an eigenvalue of 2.90, accounting for 72.56% of the variance and factor loadings ranging from 0.78 to 0.90. The Cronbach's coefficient alpha for this scale is 0.87.

To develop measures for *dynamic capabilities*, we adopted items used in previous research [31], [44], [55]. They were measured using Likert scales ranging from 1, "strongly disagree" to 5, "strongly agree."

Technology capabilities were measured using three items [31], [55]. The respondents rated the following, relative to our major competitors.

- 1) The level of efficiency and effectiveness of production is higher.
- 2) The level of the use of technological experience is higher.
- 3) The level of the use of production facilities is higher.

The PCA analysis indicated that all items loaded on a single factor, with an eigenvalue of 2.06, accounting for 68.70% of the variance and factor loadings ranging from 0.81 to 0.86. The Cronbach's coefficient alpha for this scale is 0.77.

Marketing capabilities were composed of four items [34], [44], including, relative to our major competitors.

- 1) We understand the customer better.
- 2) We understand competitors better.
- 3) The level of integration of marketing activities is higher.
- 4) Skills of segment and position of market are stronger.

The PCA analysis indicated that all items loaded on a single factor, with an eigenvalue of 2.51, accounting for 62.84% of the variance and factor loadings ranging from 0.76 to 0.82. The Cronbach's coefficient alpha for this scale is 0.80.

Market-linking capabilities were composed of five items [31], [55], including, relative to our major competitors: 1) stronger customer-linking capabilities (i.e., creating and managing durable customer relationships); 2) creating stronger durable relationships with suppliers; 3) retaining customers better; 4) stronger channel-bonding capabilities (creating durable relationship with channel members); and 5) stronger relationship with members of marketing channel. The PCA analysis indicated that all items loaded on a single factor, with an eigenvalue of 3.18, accounting for 63.67% of the variance and factor loadings ranging from 0.75 to 0.83. The Cronbach's coefficient alpha for this scale is 0.86.

Risk orientation is composed of two items [68], including.

- 1) Decision makers prefer high-risk projects which have a chance for high returns.
- 2) Firm's operations are considered high-risk.

The PCA analysis indicated that all items loaded on a single factor, with an eigenvalue of 1.62, accounting for 81% of the

variance and factor loadings both are 0.90. The Cronbach's coefficient alpha for this scale is 0.76.

We included a number of variables to control for additional influences. *Firm scale*, measured as the total sales in 2007, was controlled because it can influence organizational R&D investment [69]. *Firm age* was measured as the number of years since the formation of the firm. Industry effect was controlled because high-tech firms could face more technological opportunities and thus influence the process of strategic change. *Industry (high-tech)* was measured using a dummy variable, with 1 representing high-tech firms, and 0 represents other firms [69]. *Organizational slack* has significant effect on firms, and is measured by asking firms to indicate whether they can provide enough financial resources to business activities. Further, we control for *environmental dynamism* [7], measured by asking firms to indicate the extent of environment change they faced.

Since our constructs were all measured using a single instrument, there is the potential of common method bias. We took measures to minimize its occurrence. First, the questionnaire items were randomly ordered. Second, the survey items did not appear as having right, wrong, or socially desirable answers. We further tested for the occurrence of the bias in our results via Harman's one-factor test and a confirmatory factor analysis (CFA) analysis [70]. If substantial common method bias exists, either a single factor will emerge or one general factor will account for most of the variance [70]. The first factor explained 32.81% of the variance, and no general factor was apparent in the unrotated factor solution. Second, we used a CFA to test for common method variance [71]. A model positing that a single factor underlies the study variables was assessed by linking all items of the dependent and independent factors to a single factor. This model did not fit the data well (chi-square/df = 6.96, RMSEA = 0.17, CFI = 0.77, NFI = 0.74, IFI = 0.77, and GFI = 0.62). When the common factor was removed and all items were assigned to their theoretical factors, the model fit the data well (chi-square/df = 2.33, RMSEA = 0.077, CFI = 0.95, NFI = 0.91, IFI = 0.95, and GFI = 0.84). Thus, there is no evidence of common method bias.

IV. RESULTS

Descriptive statistics are shown in Table I. We used the optimal scaling regression (OSR) model to test our hypotheses because of the categorical nature of our data. The ordinary least square (OLS) regression cannot capture the uneven distribution of categorical variables; that is, the distance between two consecutive numbers may not be equal [72]. This attribute of categorical variables can be appropriately addressed by the application of the OSR model, because the space between categories accepts an optimized quantification [72]. This method has been successfully used in other management research (e.g., [73]). We also conducted a robustness test by applying the OLS model, results of which were weaker¹; these results suggest that the OSR model provides a more appropriate fit for our model.

Table II reports the results of the regression analyses testing the main and moderating effects. *Hypothesis 1* predicted a

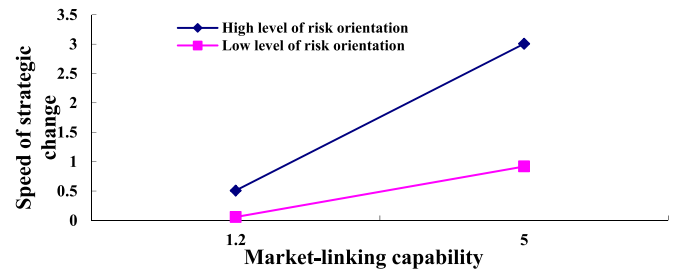


Fig. 2. Moderating effect of risk orientation.

positive relationship between technological capabilities and the speed of strategic change. As shown in model 2, the relationship between technology capabilities and the speed of strategic change is positive and statistically significant ($\beta = 0.35$, $p < 0.001$). *Hypothesis 1*, therefore, receives support. *Hypothesis 2* predicted a positive relationship between marketing capabilities and the speed of strategic change. The coefficient for marketing capability suggests a statistically significant positive effect ($\beta = 0.32$, $p < 0.001$), thus providing support for *Hypothesis 2*. *Hypothesis 3* predicted a positive relationship between market-linking capabilities and the speed of strategic change. The coefficient for market-linking capability is statistically significant positive ($\beta = 0.14$, $p < 0.05$), providing support for *Hypothesis 3*.

We also hypothesized a moderating effect of risk orientation on the effect of technological capabilities (*Hypothesis 4*), marketing capabilities (*Hypothesis 5*) and market-linking capabilities (*Hypothesis 6*) on the speed of strategic change. In model 3, the coefficient for the interaction of technological capabilities and risk orientation is negative ($\beta = -0.13$, $p < 0.05$). Thus, *Hypothesis 4* is not supported. The coefficient for the interaction of marketing capability and risk orientation is negative ($\beta = -0.13$, $p < 0.01$). *Hypothesis 5* is, therefore, not supported. However, the coefficient for the interaction of market-linking capability and risk orientation is positive and statistically significant ($\beta = 0.15$, $p < 0.01$). This result is plotted in Fig. 2. As shown in the plot, the relationship between market-linking capabilities and speed of strategic change is stronger for more risk-taking managers than for more risk-averse managers, thus providing support for *Hypothesis 6*.

Finally, *Hypothesis 7* predicted an inverted-U relationship between the speed of strategic change and performance. As illustrated in model 5, the speed of strategic change has a statistically significant positive effect on firm performance ($\beta = 0.13$, $p < 0.05$). As shown in model 6, while the first order term of the speed of strategic change is still significant ($\beta = 0.12$, $p < 0.10$), its squared value has a significant and negative effect ($\beta = -0.13$, $p < 0.10$), providing support for the inverted-U relationship in *Hypothesis 7*.

V. DISCUSSION

Despite the consensus acknowledgment of the importance of resources and capabilities to strategic change within the management literature [2], the broader question of how resources and capabilities affect strategic change itself remains unresolved. We propose that our understanding of the relationship

¹Results are available from the authors.

TABLE I
DESCRIPTIVE STATISTICS AND CORRELATIONS ^a

Variables	Means	s.d.	1	2	3	4	5	6	7	8	9	10	11
1. Organization scale	10.70	2.92	-										
2. Organization age	2.39	0.94	0.36**										
3. Industry (Hi-tech)	0.39	0.49	0.05	0.01									
4. Slack	3.11	1.23	0.08	-0.05	0.14*								
5. Environment dynamism	4.00	0.91	0.10	-0.04	-0.02	0.03							
6. Technological capability	3.55	0.88	0.05	-0.03	0.15*	0.22**	0.04						
7. Marketing capability	3.67	0.83	0.13	-0.06	0.08	0.33**	0.16*	0.49**					
8. Market-linking capability	3.86	0.78	0.02	-0.07	0.07	0.32**	0.13	0.46**	0.66**				
9. Risk	2.56	1.06	0.00	0.00	0.02	-0.00	0.01	-0.06	-0.07	-0.09			
10. Speed of strategic change	3.51	0.92	-0.05	-0.18*	-0.04	0.17*	0.07	0.46**	0.47**	0.44**	-0.06		
11. Performance	3.74	0.87	0.16*	-0.05	0.14	0.25**	0.01	0.29**	0.36**	0.24**	0.01	0.17*	

^a: N = 213; ***p* < 0.01; **p* < 0.05.TABLE II
RESULTS OF THE REGRESSION

Variable	Speed of strategic change			Performance		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Control Variables						
Organization scale	-0.02 (.07)	-0.15** (.06)	-0.15** (.06)	0.22** (.07)	0.23** (.07)	0.23** (.07)
Organization age	-0.18* (.07)	-0.11† (.06)	-0.10† (.06)	-0.12 (.07)	-0.12** (.07)	-0.12† (.07)
Industry (Hi-tech)	0.00 (.07)	-0.05 (.06)	-0.04 (.06)	0.11 (.07)	0.12† (.07)	0.13* (.07)
Slack	0.411*** (.067)	-0.10† (.06)	-0.09 (.06)	0.22*** (.07)	0.23*** (.07)	0.24*** (.08)
Environment dynamism	0.084 (.066)	-0.05 (.06)	-0.06 (.06)	-0.13* (.07)	-0.13*** (.07)	-0.12* (.07)
Direct Effects						
Technological capability		0.35*** (.07)	0.29*** (.07)	0.19** (.08)	0.16** (.08)	0.15** (.07)
Marketing capability		0.32*** (.07)	0.34*** (.07)	0.12† (.08)	0.09 (.08)	0.09 (.08)
Market-linking capability		0.14* (.07)	0.12† (.07)	0.13† (.08)	0.13† (.08)	0.15* (.08)
Risk orientation			-0.08 (.06)	0.10† (.07)	0.09 (.07)	0.10 (.07)
Speed of strategic change					0.13* (.07)	0.12† (.08)
Speed of strategic change ²						-0.13† (.08)
Interactions						
Technology capability × Risk orientation			-0.13* (.07)			
Marketing capability × Risk orientation			-0.13** (.07)			
Market-linking capability × Risk orientation			0.15** (.06)			
R ²	0.22	0.45	0.49	0.30	0.32	0.32
Adjusted-R ²	0.18	0.40	0.40	0.22	0.22	0.23
Δ R ²		0.23	0.04		0.02***	0.01***
F	5.60***	8.34***	5.13***	3.41***	3.24***	3.49***

****p* < 0.001, ***p* < 0.01, **p* < 0.05, †*p* < 0.10 (two-tailed test); standard errors are in parenthesis.

N = 213.

between dynamic capabilities and strategic change can be improved through the investigation of technological, marketing, and market-linking capabilities, their impact on the speed of strategic change, and subsequent effect on firm performance.

Specifically, we find the following.

- 1) Technological, marketing, and market-linking capabilities facilitate rapid strategic change.
- 2) Risk orientation of decision-makers strengthens the market-linking—speed of strategic change relationship.
- 3) The speed of strategic change has an inverted-U relationship with organizational performance.

Our findings suggest that firms not only need to adjust strategies to match the changing environment, but also have corresponding capabilities to enable fast strategic change.

A. Theoretical Implications

These findings provide several contributions to the dynamic capabilities literature. First, while it is acknowledged that ca-

pabilities affect actions, this study enriches our understanding of strategic change by examining a hitherto unexamined dimension of strategic change: its speed. While most prior research on strategic change has focused on the magnitude of change, the speed with which strategic change is implemented can also affect performance. Increasingly competitive environments necessitate firms to learn and act faster [5] and seize opportunities to gain competitive advantages before competitors [19]. A firm's portfolio of capabilities facilitates the speed of implementing strategic change. While faster implementation of strategic change has a positive effect on performance, strategic change implemented too quickly rather yields negative performance repercussions.

Furthermore, despite the prominence of the concepts of strategic change and flexibility in the literature on dynamic capabilities, scant research to date has empirically examined the link between dynamic capabilities and the speed of strategic change. This study investigates the unexamined question of “*how*” dynamic capabilities influence the speed of strategic change [2],

[27]. Our findings provide more interesting implications on differing roles for various capabilities in the process of strategic change.

Lastly, findings of this study reveal that managerial risk orientation influences the relation between dynamic capabilities and the speed of strategic change in different ways. While previous research focused on how decision makers' characteristics directly affect a firm's strategic change [14], this research differentiates itself from prior studies by testing the indirect effect of managers' risk orientation. Specifically, we examined how the risk preference of managers influences the firm's utilization of dynamic capabilities. We found that managerial risk orientation moderates the relationship between dynamic capabilities and the speed of strategic change, but not in a singular direction.

B. Managerial Implications

Our arguments and findings also provide practical implications to managers, especially those whose firms are located in dynamic environments. Our findings suggest that a firm's portfolio of dynamic capabilities facilitates the speed with which it can carry out strategic change, but does so in different ways. Thus, managers need to understand the roles played by different capabilities in the process of strategic change and cultivate them purposefully to enable planned strategic changes. Moreover, managerial risk orientation will influence the relation between dynamic capabilities and the speed of strategic change in divergent ways. Managers should be aware of their own risk orientation when making decisions centered on strategic changes. Furthermore, how quickly a firm can initiate and implement strategic change does not improve firm performance in a linear manner. Although the faster implementation of strategic change generally improves performance, strategic changes initiated and implemented too quickly will lead to negative performance repercussions. Thus, managers need to control the pace of strategic change to reap performance benefits.

C. Limitations and Future Research

This study's limitations indicate opportunities for future research. First, contrary to our predictions, we found that managerial risk orientation has a negative moderating effect on the relationships between technological and marketing capabilities and the speed of strategic change. Specifically, more risk-taking managers made slower strategic changes the more technological and marketing capabilities their firms had. One possible explanation for this counterintuitive finding is that in a transition economy such as China, from which our sample is drawn, fast strategic change may actually be the norm. The dynamism created by changes to the economy necessitates firms to constantly change their strategy. The riskier option may be slower or no strategic change. Specifically, as a firm has more technological and marketing capabilities that help buffer the firm from the environment, more risk-taking managers may actually deviate from the norm and rather make and implement strategic change at a slower pace.

This occurs for technological and marketing capabilities, but not for market-linking capabilities, because technological and marketing capabilities differ from market-linking capabilities: The former are contained within the organization, while the latter exist in relationships with external stakeholders [31]. When high value stakeholders that make up these market-linking relationships such as customers change, or need change, even the most reticent managers are forced to change to maintain the relationship. Technological and marketing capabilities, on the other hand, are internal and thus entirely within managerial discretion. Thus, future research can further elucidate the role of the environment on the effect of managerial risk orientation on the capabilities-strategic change relationship in different economies and industry environments. Whether this result is unique to the Chinese context, or applies more broadly to other dynamic markets is clearly an issue for future research.

Second, our results provide support for an inverted-U relationship between the speed of strategic change and performance. We find that, beyond a point, faster strategic change rushed decisions and negatively impacted performance. However, because of the multiindustry nature of our sample, we were not able to fully investigate the dynamics associated with this nonlinear relationship. For example, at what critical threshold strategic change speed does performance reverse? Furthermore, we suspect this threshold will depend on firm and industry factors. Future research can examine how this speed threshold varies across firms and environments and how firms control the speed of strategic change.

Lastly, because firms in our sample are drawn from different industries and thus no singular environment, we measure strategic change speed subjectively. This approach enables us to measure the speed of strategic change as it occurs during the normal course of business. Prior studies that utilized objective measures of the speed of strategic change [6], [9] were able to do so because, as single industry studies, firms in their samples were exposed to the same tractable stimulus that created a need for strategic change. Future research can investigate the feasibility of utilizing objective measures of speed in a multiindustry sample.

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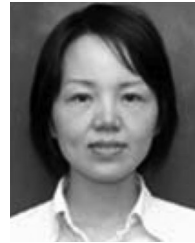
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